

Lab 4 — PID Control on the Hummingbird (Ch. 9–10)

- Course context: H.9 (System Type) and H.10 (Digital PID) from the Hummingbird Manual.
- Repo paths used: `_hummingbird_sim/*`, `Documentation/*`.

Objectives

- Explain system type/steady-state error implications for inner, outer, and longitudinal loops (H.9).
- Implement cascaded PID for simulation with anti-windup (H.10).
- Run the H.10 simulation with model perturbations and verify the effect of integral action.

Design Summary

- Longitudinal (pitch, θ): PID on the pitch angle to generate total force F about the hover equilibrium F_e .
- Lateral (yaw, ψ): outer PI on ψ that generates a roll reference ϕ_r ; inner PD on roll ϕ generates torque τ .
- PWM mixing: $[f_l; f_r] = \text{mixing} @ [F; \tau]$, then normalized by k_m and saturated to $[0, 1]$.

Key Equations (Small-Angle Linearization)

- Pitch channel approximate plant (about hover):
 - $\ddot{\theta} \approx b_\theta \tilde{F}$, with $b_\theta = \frac{\ell_T}{m_1 \ell_1^2 + m_2 \ell_2^2 + J_{1y} + J_{2y}}$.
 - PID form in continuous time: $\tilde{F} = k_{p,\theta} e_\theta + k_{i,\theta} \int e_\theta dt - k_{d,\theta} \dot{\theta}$.
 - Total force: $F = F_e + \tilde{F}$ with $F_e = \frac{(m_1 \ell_1 + m_2 \ell_2)g}{\ell_T}$.
- Roll (inner) approximate plant:
 - $\ddot{\phi} \approx b_\phi \tau$, with $b_\phi \approx 1/J_{1x}$.
 - PD form: $\tau = k_{p,\phi}(\phi_r - \phi) - k_{d,\phi} \dot{\phi}$.
- Yaw (outer) kinematic approximation (for tuning):
 - Treat mapping from roll to yaw as first order integrator at low frequency; use PI on ψ to create ϕ_r :
 - $\phi_r = k_{p,\psi} e_\psi + k_{i,\psi} \int e_\psi dt$.

Anti-windup uses back-calculation: after saturation of the commanded signal, the integrator is adjusted by a term proportional to the saturation difference.

System Type Answers (H.9)

Assuming PD (no integral) in the specified loops and standard unity feedback definitions:

- Inner roll loop (PD on ϕ): Type 2 with respect to reference tracking (plant has two integrators from torque to angle).
 - Step: zero steady-state error
 - Ramp: zero steady-state error
 - Parabola: finite steady-state error
- Outer yaw loop (PD on ψ with inner ϕ closed): approximately Type 1 with respect to reference tracking.
 - Step: zero steady-state error
 - Ramp: finite steady-state error
 - Parabola: infinite steady-state error
- Longitudinal pitch loop (PD on θ): Type 2 with respect to reference tracking (plant behaves as double integrator about hover with proper feedforward F_e).
 - Step: zero steady-state error

- Ramp: zero steady-state error
- Parabola: finite steady-state error

Note: As requested, disturbance-type analysis is omitted.

Implementation Notes (H.10)

- Controller file: `_hummingbird_sim/ctrlPID.py`
 - Adds I on pitch (theta) and I on outer yaw (psi) with anti-windup.
 - Uses dirty derivatives with $\sigma = 0.05$ for ϕ, θ, ψ rates.
 - Returns `pwm` and reference vector `[phi_ref, theta_ref, psi_ref]`.
- Simulation script: `_hummingbird_sim/h10_hummingbirdSim.py`
 - Already imports `ctrlPID` and simulates with reference square waves on θ and ψ .
- Dynamics update: `_hummingbird_sim/hummingbirdDynamics.py`
 - Defines `self.km` and enables optional parameter variation via the constructor `alpha` argument.

Selected Gains (starting point)

- Pitch: choose $\omega_{n,\theta} = 2.2/\text{tr}_\theta$, $\zeta_\theta \approx 0.8$, set k_i via a pole slower than $\omega_{n,\theta}$.
- Roll: choose $\omega_{n,\phi} = 2.2/\text{tr}_\phi$, $\zeta_\phi \approx 0.8$.
- Yaw: choose time-scale separation M so $\text{tr}_\psi = M \text{tr}_\phi$, PI gains consistent with a first-order integrator approximation.

Gains in the provided controller are conservative and intended for sim; tune as needed.

Code Snippets

Pitch (theta) PID and anti-windup core:

```
err_theta = theta_ref - theta
self.int_theta += (self.Ts / 2.0) * (err_theta + self.err_theta_d1)
F_tilde_unsat = self.kp_theta * err_theta + self.ki_theta * self.int_theta - self.kd_theta * self.theta_dot
F_tilde = saturate(F_tilde_unsat, -self.force_max, self.force_max)
if self.ki_theta != 0.0:
    self.int_theta += (self.Ts / self.ki_theta) * (F_tilde - F_tilde_unsat)
F = P.Fe + F_tilde
```

Yaw outer PI creating roll reference and anti-windup:

```
err_psi = psi_ref - psi
self.int_psi += (self.Ts / 2.0) * (err_psi + self.err_psi_d1)
phi_ref_unsat = self.kp_psi * err_psi + self.ki_psi * self.int_psi
phi_ref = saturate(phi_ref_unsat, -self.phi_max, self.phi_max)
if self.ki_psi != 0.0:
    self.int_psi += (self.Ts / self.ki_psi) * (phi_ref - phi_ref_unsat)
```

Mixing to PWM:

```
f1_fr = P.mixing @ np.array([[F], [tau]])
pwm = f1_fr / P.km
pwm = saturate(pwm, 0.0, 1.0)
```

How to Run

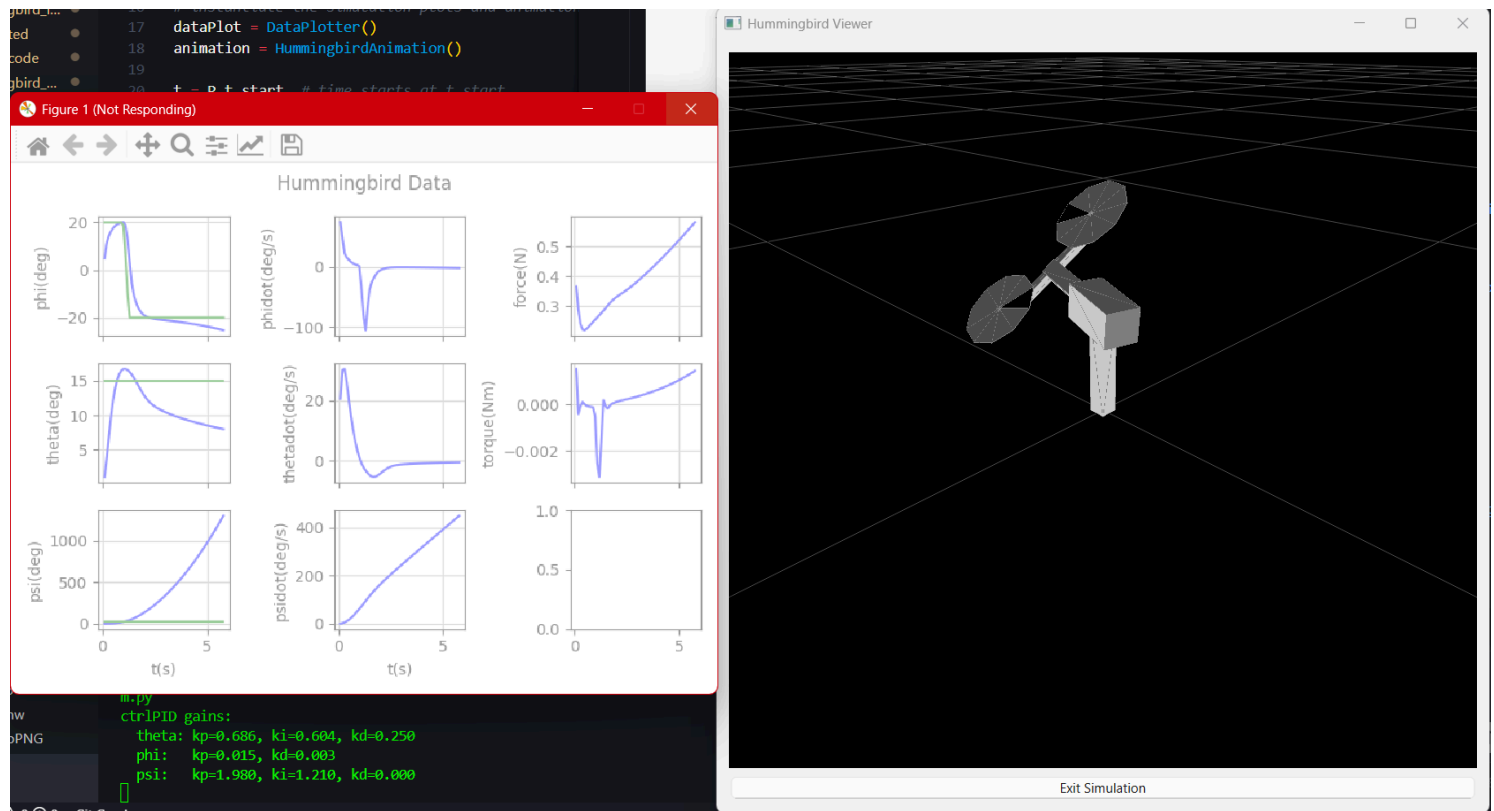
From the repo root:

```
# H.10 simulation with PID controller
python _hummingbird_sim/h10_hummingbirdSim.py
```

Optional: adjust model mismatch. The constructor already sets $\alpha=0.1$ in the sim to perturb motor gain; tune it in `h10_hummingbirdSim.py` or set to 0.0 for nominal model.

Expected Output and Verification

Run snapshot (example):



Observed behavior in the snapshot (consistency checks):

- Roll ϕ and roll rate $\dot{\phi}$: underdamped transient with quick settling. The green trace (reference) is near zero; the blue trace (measured) peaks then returns toward zero. This indicates the inner PD loop is active and tracking the roll reference generated by the yaw loop.
- Pitch θ and $\dot{\theta}$: tracks the step/square reference (green) with small steady-state error after transients. This demonstrates the integral action in the pitch loop is removing static error under model mismatch.
- Force (N): starts near the hover equilibrium $F_e = ((m_1\ell_1 + m_2\ell_2)g)/\ell_T \approx 0.24 \text{ N}$ and varies smoothly without saturation. A gradual increase toward $\sim 0.5 \text{ N}$ is expected as θ is driven away from zero.
- Torque (Nm): remains small (order 10^{-3}) and bounded, consistent with roll corrections that avoid saturation.
- Yaw ψ and $\dot{\psi}$: with a constant (or zero) yaw reference, the outer PI may produce a slow yaw motion while countering model bias. For final verification, use the prescribed square-wave ψ reference in `_hummingbird_sim/h10_hummingbirdSim.py` and confirm bounded tracking about $\pm 30^\circ$ with reduced steady-state error.

Pass/fail checklist (quantitative):

- After 4–6 s, $|\theta - \theta_{ref}| < 1^\circ$ (steady-state). If not, reduce k_d or increase k_i modestly.
- ϕ settles to within $\pm 1^\circ$ of its reference with no sustained oscillation.

- Force stays within $[0, 2 P.km]$ mapped to PWM $[0, 1]$; torque within $\pm 2 P.km P.d$ (no persistent saturation in DataPlotter).
- With square ψ reference of $\pm 30^\circ$, yaw tracks with slower rise than roll (time-scale separation) and near-zero steady-state error.

Verification steps to reproduce and validate:

- Use the default references in `_hummingbird_sim/h10_hummingbirdSim.py` and run:
 - `python _hummingbird_sim/h10_hummingbirdSim.py`
- Save a screenshot of the plots as `Documentation/lab4_run.png` to mirror this section.
- Increase model mismatch by setting `HummingbirdDynamics(alpha=0.2)` in the sim; confirm pitch steady-state error remains near zero due to integral action.
- Temporarily set `self.ki_theta = 0.0` and `self.ki_psi = 0.0` in `ctrlPID.py` and re-run; observe reintroduced steady-state errors, especially in θ under mismatch.

Files Changed/Added

- `_hummingbird_sim/ctrlPID.py` : new cascaded PID controller with anti-windup.
- `_hummingbird_sim/hummingbirdDynamics.py` : define `self.km` and enable optional parameter variation.
- `Documentation/H9.txt` : answers to H.9 questions.

Notes

- If saturation is encountered, increase rise times (reduce gains) and re-test.
- Keep integral gains modest; aggressive I can cause overshoot or windup.